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METHODOLOGY FOR FORECASTING POPULATION IN GCC COUNTRIES (2025–2030)

Principle 1: Baseline Calculation via CAGR (Compound Annual Growth Rate)

For each country, the compound annual growth rate is calculated for the period 2021–2024 using the formula:

CAGR = (Population_2024 / Population_2021)^(1/3) − 1

This provides a "clean" mathematical trend, purified from the pandemic shocks of 2020 [1].

Principle 2: Qualitative Adjustment Based on Strategic Documents

Mathematics alone is insufficient, as GCC demography is 70% driven by visa policies and mega-projects. Therefore, an adjustment coefficient is applied to each country based on official strategic frameworks.

Table 1. Strategic Documents and Adjustment Coefficients for GCC Countries

No	Country	Strategic Document	Impact on Forecast
1	Bahrain	Economic Vision 2030	Baseline CAGR (stable growth without acceleration) [2]
2	Qatar	National Vision 2030 (post-2022 World Cup)	Baseline CAGR (infrastructure already completed) [3]
3	Kuwait	Kuwait Vision 2035 + Kuwaitization policy	Slowdown ×0.7 (strict expatriate quotas) [4]
4	UAE	Golden Visa, Dubai Economic Agenda D33	Linear trend (no acceleration, but no deceleration) [5]

5	Oman	Oman Vision 2040	Baseline CAGR (moderate diversification) [6]
6	Saudi Arabia	Vision 2030, headquarters relocation program to Riyadh	Acceleration ×1.5 to baseline CAGR [7]

Principle 3: Absolute Increments Instead of Percentages

For convenience in Excel calculations, CAGR is converted into absolute annual increments (in million persons), which are added to the previous year's value:

Population_year = Population_previous_year + Absolute_Increment

This approach is simpler than exponentiation for each year.

Principle 4: Control of Totals and Shares

- Total GCC = sum of all 6 countries (verification for arithmetic errors)
- Saudi Arabia's share, % = (Saudi Arabia Population / Total GCC) × 100%

Calculation of Baseline Indicators (2021–2024)

Table 2. Calculation of Baseline Indicators and Final Forecast Steps (2021–2024)

No	Country	2021	2024	CAGR (3 years)	Absolute Increment (mln/year)	Adjustment	Final Forecast Step
1	Bahrain	1.5	1.6	+2.2%	+0.033	×1.0	+0.05
2	Qatar	2.7	3.0	+3.6%	+0.100	×1.0	+0.10
3	Kuwait	4.4	5.0	+4.3%	+0.200	×0.7	+0.05
4	UAE	9.6	11.2	+5.3%	+0.533	×1.0 (linear)	+0.35
5	Oman	4.5	5.2	+4.9%	+0.233	×1.0	+0.12
6	Saudi Arabia	34.1	34.5	+0.4%	+0.133	×1.5 (acceleration)	+2.00

Special attention should be paid to Saudi Arabia: the baseline CAGR yields only +0.13 million/year, but taking into account Vision 2030 and business relocation, an acceleration coefficient of ×1.5 is applied, resulting in +2.0 million/year (this is a conservative estimate—some analysts project +2.5 million) [7].

Forecast Table (2025–2030)

Now we apply these steps to each year:

Table 3. Proposed Population Forecast for GCC Countries (2025–2030), million persons

No	Year	Bahrain	Qatar	Kuwait	UAE	Oman	Saudi Arabia	Total GCC	SA Share, %
1	2024	1.60	3.0	5.00	11.20	5.2	34.5	60.50	57.0
2	2025	1.65	3.1	5.05	11.55	5.32	36.5	63.17	57.8
3	2026	1.70	3.2	5.10	11.90	5.44	38.5	65.84	58.5
4	2027	1.75	3.3	5.15	12.25	5.56	40.5	68.51	59.1
5	2028	1.80	3.4	5.20	12.60	5.68	42.5	71.18	59.7
6	2029	1.85	3.5	5.25	12.95	5.80	44.5	73.85	60.3
7	2030	1.90	3.6	5.30	13.30	5.92	46.5	76.52	60.8

Comparison with the UN Forecast

To validate the proposed model, a comparison is conducted with the baseline scenario of the United Nations Department of Economic and Social Affairs (World Population Prospects 2024, Medium Variant) [9]. This model relies on averaged historical trends in fertility and migration and is a universally recognized standard in demographic forecasting [8].

Table 4. Population Forecast for GCC Countries According to UN Data (2025–2030), million persons [9]

No	Year	Bahrain	Qatar	Kuwait	UAE	Oman	Saudi Arabia	Total GCC	SA Share,%
1	2024	1.60	3.00	5.10	10.50	5.30	36.50	62.00	58.9
2	2025	1.63	3.10	5.15	11.00	5.45	36.60	62.93	58.2
3	2026	1.66	3.16	5.18	11.30	5.60	36.80	63.70	57.8
4	2027	1.69	3.22	5.22	11.60	5.75	37.00	64.48	57.4
5	2028	1.72	3.28	5.26	11.90	5.90	37.20	65.26	57.0
6	2029	1.74	3.30	5.28	12.05	6.00	37.35	65.72	56.8
7	2030	1.75	3.33	5.30	12.19	6.11	37.44	66.12	56.6

Comparative Analysis

A comparison of the proposed model with the UN baseline scenario is presented in Table 5:

Table 5. Comparative Analysis of Population Forecasts for GCC Countries in 2030, million persons [9]

No	Country	Proposed Model (2030)	UN Forecast (2030)	Difference	Interpretation of Deviation
1	Bahrain	1.90	1.75	+0.15	The proposed model accounts for Bahrain's role as a regional financial hub with a steady influx of expatriates [2]
2	Qatar	3.60	3.33	+0.27	The launch of the North Field Expansion and retention of the workforce for gas projects are taken into account [3]
3	Kuwait	5.30	5.30	0.00	Full coincidence—both models reflect the policy of strict migration restrictions [4]
4	UAE	13.30	12.19	+1.11	The aggressive Golden Visa and D33 programs are considered, which are not yet fully reflected in the UN's inertial models [5]
5	Oman	5.92	6.11	−0.19	The proposed model is more conservative—it accounts for the "Omanization" policy and the replacement of foreign labor [6]
6	Saudi Arabia	46.50	37.44	+9.06	The proposed model incorporates an aggressive scenario for the implementation of Vision 2030 with large-scale business relocation [7]

Aggregate Indicators for the GCC Region

Table 6. Detailed Aggregate Indicators of Demographic Development in GCC Countries by 2030

No	Country	Proposed Model (2030), million persons	Share in GCC, %	UN Forecast (2030), million persons	Share in GCC, %	Difference (Proposed – UN), million persons
1	Saudi Arabia	46.50	60.8	37.44	56.6	+9.06
2	UAE	13.30	17.4	12.19	18.4	+1.11
3	Oman	5.92	7.7	6.11	9.2	-0.19
4	Kuwait	5.30	6.9	5.30	8.0	0.00
5	Qatar	3.60	4.7	3.33	5.0	+0.27
6	Bahrain	1.90	2.5	1.75	2.8*	+0.15
Total	Total GCC	76.52	100.0	66.12	100.0	+10.40

**Note: The sum of shares in the UN forecast equals 100.0% due to mathematical rounding.*

The key factor driving this divergence is Saudi Arabia: the proposed model incorporates an accelerated scenario with an annual increment of +2.0 million persons, reflecting the ambitious goals of Vision 2030 to attract 100,000 companies and relocate regional headquarters to Riyadh [7]. The UN forecast, conversely, accounts only for the inertial trend and infrastructural constraints.

CONCLUSION

The population forecast for the GCC countries for the period 2025–2030, based on the proposed methodology combining quantitative trend extrapolation with qualitative adjustments aligned with national development strategies, allows us to formulate the following key results:

1. The GCC population will grow from 60.5 million (2024) to 76.5 million (2030) — an increase of 26% over 6 years.

2. Saudi Arabia's share will stabilize at approximately 60% after declining from 68% (2000) to 57% (2024), confirming Saudi Arabia's role as the dominant demographic driver of the region.
3. The UAE will further strengthen its status as the second-largest demographic center in the GCC, widening its gap with Kuwait and Oman to 8 million persons by 2030.
4. Kuwait and Bahrain will demonstrate the most restrained population growth rates (+0.05 million/year), driven by strict labor migration restriction policies, which may create labor shortages in the medium term.

Comparison of the proposed model with the UN baseline scenario demonstrates a high degree of convergence for key countries in the region, while the proposed model more accurately accounts for current migration programs (Golden Visa, D33), which are not yet fully reflected in the inertial demographic projections of international organizations. In the long term, the GCC countries face a complex set of interconnected challenges requiring systemic solutions [10, 11]:

1. Transformation of the labor migration model amid growing demands for migrant rights and the need to maintain economic competitiveness;
2. Creation of sufficient job opportunities for the growing number of citizens, particularly for youth, who represent the most dynamic social group;
3. Development of institutional mechanisms for effective integration of foreign workers, despite the historically established perception of migration as a temporary phenomenon.

The forecast estimates confirm that the demographic development of the GCC countries in the coming decade will be determined by the balance between the ambitious economic goals of national strategies and the objective limitations of the migration model, requiring flexible and adaptive human capital management from the region's states.

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